

In 2011, “The Fundamental Law of Road Congestion: Evidence from US Cities” was published. My lived experience of this paper and its consequences has most frequently come in the form of being told “adding a lane doesn’t actually reduce traffic,” but that’s not really true.

The core idea is induced demand - as a road becomes less “costly” to drive on (it’s quicker) more people will drive on it, driving up cost (in time), until eventually it reaches an equilibrium where the time cost is the same as it was before. Hence, no long-term benefit.

The extent to which this happens is variable and depends largely on the population of the area. If there are more people nearby who can elect to drive, change when they drive, change where they drive, etc. traffic will be more responsive to road capacity. It also depends on the actual road structure, constructing additional lanes in bottleneck areas is very effective at improving the flow of traffic.

Now let’s look at what the paper actually says:

For interstate highways in metropolitan areas we find that VTK (vehicle-kilometers travelled) increases one for one with interstate highways, confirming the “fundamental law of highway congestion” suggested by Anthony Downs [he wrote *Stuck in Traffic*, the seminal work in this field]

You’ll notice that this is about interstate highways, which are, for reasons which should be relatively apparent, the roads that are most responsive to this effect.

They then note:

We also uncover **suggestive evidence** that this law may extend beyond interstate highways to a broad class of major urban roads, a “fundamental law of road congestion.”

I find their convention of naming everything a fundamental law to be a bit weird, but more importantly, this “suggestive evidence” is an

estimated roadway elasticity of VKT between 0.67 and 0.89, depending on the decade in OLS. Changes in the boundaries of urban areas over time and the weakness of our instruments for this class of roads preclude reliable first-difference and IV estimates.

“Depending on the decade in OLS” is not a good sign (for a number of reasons), and leaves an open question regarding the validity of this result. And their alternative specifications (first-diff and IV) are, as they mentioned, also not reliable. And, it’s still about major urban roads, and a substantially lower, wider, and less confident estimate.

Additionally, VTK does not equal congestion. They explicitly state that their causal inference is about the impact of lanes (1% capacity increase) on the amount of driving (1% VTK increase), not the amount of congestion, and they do no work in the paper to measure congestion directly.

I find it un-intuitive that actual realized congestion would scale in this way with the ratio of lane capacity and VTK.

In summary, I’m not actually very informed on this subject but I think the people I’ve talked to about this have vastly overstated the scale, speed, and confidence of these findings.